Large Language Models against Historical Ciphers: A Profiled Record of 483 Targets, 1398–1945

Anonymous submission

Abstract

We report what agentic large language models (LLMs) achieved on historical cipher targets from any region, dated 1398–1945 (mostly European diplomatic letters, with 33 targets from North America, Latin America, East Asia and the Middle East), attacked between 14 September 2026 and 2 November 2026. Each target has a fixed profile: the cipher system, the documents, what the model started from, whether a reading already existed, the ordered solution steps, and an outcome split into *method* (how the text was obtained) and *extent* (complete or partial). The text was obtained for 320 of the 434 targets that could be worked (74 %), but that figure hides the methods. The key was recovered here for 102: 43 from the ciphertext alone, 47 from plaintext beside the cipher and 12 from plaintext found elsewhere. For 114 a key that already existed was found and applied. For 104 a decipherment already existed. From ciphertext alone the model recovered the key in 37 of 110 attempts. On the 129 targets with no known earlier reading, the text was obtained for 64. The model's main strength was locating keys, cribs and decipherments; its main failures were noisy transcriptions, short texts and large nomenclators without key material.

1 Introduction

Historical cryptology has been reshaped twice in the last two decades: by metaheuristic search, which made ciphertext-only attacks on homophonic and transposition ciphers routine (Lasry, 2018), and by shared collections such as DECODE, which put thousands of ciphertexts and keys in one place (Megyesi et al., 2019; Megyesi et al., 2020). Large language models add a third element. An agentic model can read a manuscript image, search archive catalogues and printed editions, write and run a solver, and judge whether its output is language. Whether that adds up to useful cryptanalysis, and on which ciphers, is an empirical question.

This paper answers it from a working record rather than a benchmark. Between 14 September 2026 and 2 November 2026 agentic models worked through lists of unsolved or unread historical ciphers: Tomokiyo's list of unsolved historical ciphers (Tomokiyo, 2026), Schmeh's list of the top fifty unsolved encrypted messages (Schmeh, 2017), the DECODE records marked non-decrypted, and a sweep of the manuscript catalogue of the Bibliothèque nationale de France. All of the work was done in one version-controlled repository, and that repository is the record we analyse: a folder per target holding the agent's working notes, the transcriptions, the solver code and the readings, and a commit history (4,053 commits over 50 days) that dates every step and names the model that made it (Section 4). The weakness of such a record is that each target's notes describe its cipher in their own words. To make the targets comparable, we defined a fixed profile and filled it for every target from the notes and the files, measuring counts wherever they could be measured.

A record of this kind also forces a question the field has left informal: what counts as a success. A letter can be "deciphered" because the key was rebuilt from nothing, because a key sheet was found in another volume, or because a contemporary clerk had already written the plaintext between the lines. We therefore classify every outcome by method, following the categories proposed by the second author, and report the extent of the text obtained separately.

Our contributions are:

- a profile schema for historical cipher challenges and their solutions, in which the factors that plausibly affect success are recorded the same way each time (Section 3);
- profiles for 483 targets worked by the project, with 1117 ciphertext documents from 1398 to 1945 in 17 plaintext languages (Section 5);
- an analysis of outcome method and extent by starting point, prior reading, cipher family and length (Section 6);
- six case studies, one per outcome route, and a discussion of the failures (Sections 7–8).

2 Related work

Ciphertext-only attacks on historical ciphers by hill climbing and simulated annealing are reviewed by Lasry (2018), whose methodology covers homophonic, transposition and machine ciphers; recent HistoCrypt and *Cryptologia* papers apply it to single nomenclators (Lasry, 2023) and to whole correspondences (Lasry et al., 2023). The scoring function in such attacks is an n-gram language model, and Megyesi et al. (2023) show that models built from period texts improve decipherment of seventeenth-century and older homophonic ciphers. The DECODE database (Megyesi et al., 2019; Megyesi et al., 2020) standardises the metadata of ciphertexts and keys (provenance, dating, symbol set, transcription); our profile borrows its document fields and adds fields about the attack and its result.

Neural models have been trained to decipher directly. Sequence-to-sequence models learn to crack simple substitution ciphers (Aldarrab and May, 2021), and a causal language model trained on symbol recurrences recovers keys of real homophonic ciphers (Kambhatla et al., 2023). Bayesian inference had earlier reached 97.8 % on the Zodiac-408 cipher (Ravi and Knight, 2011). These systems are cipher solvers; they do not find sources or decide what a document is.

General-purpose LLMs have been evaluated on ciphers as reasoning tasks. CipherBank (Li et al., 2025) tests chat and reasoning models on nine encryption algorithms applied to synthetic plaintexts and finds that even the strongest models solve fewer than half of tasks that human cryptanalysts can; Maskey et al. (2025) benchmark LLMs on decrypting ciphertexts of several classical and modern schemes. Both use generated ciphertexts with known answers and no archival context. We know of no evaluation of LLMs on archival ciphertexts, where most of the work lies before and around the cryptanalysis: finding the document, transcribing it, identifying the system, and finding the key or a decipherment if one survives.

3 The profile

A profile is a JSON file in five parts, validated against a schema. Every field may take the value *unknown*; a gap is then explicit and counted, rather than filled by a plausible guess.

System. The cipher types that occur (monoalphabetic, homophonic, polyphonic, syllabic, nomenclator, code, transposition, polygraphic, polyalphabetic, superencipherment; several at once means a combined system). The symbol kind (digits, letters, graphic symbols or mixtures), digit-group width and separation, and the number of distinct symbols. Diacritics and what they do. Homophones per vowel and per consonant. The nomenclator: presence, size, kinds of entry and the number of code groups still open. Nulls, word division and doubled letters. *Key order*: whether the key assigns its signs in alphabetical order, so that a few known values predict the rest. Any sibling key from the same office.

Documents. One entry per ciphertext unit: shelfmark, date, issuing power, route, plaintext language, pages. Whether the document contains clear text, and whether a plaintext exists and where: interlinear, in the margin, on a separate folio or in a separate collection, in a printed edition, or in an unrelated source. The length in enciphered tokens and the number of distinct tokens, flagged as measured when a common script counted them from the transcription file. Who transcribed the text (the model from images, a human, a published or DECODE transcription) and the image quality.

Conditions. Whether a reading of the text existed anywhere before this project obtained it, where, whether it was found before, during or after the attempt, and whether the work used it. This is the contamination question: a model may have seen a printed decipherment in training. The inputs the attack started from and the attack class (ciphertext-only, crib, known plaintext, archive decipherment, sibling key, key from archive, published key). The human's role, the tools, the model identifiers and the dates.

Solution. The ordered steps, each with a kind (access, transcription, statistics, hypothesis, crib, solver, control, sibling key, key from source, prediction, reading, verification, literature search) and a result (worked, partial, failed, ruled out). Failed steps are recorded.

Outcome method. How the text was obtained, in one of eight categories. Three recover a key here, by cryptanalysis: *key recovered from ciphertext only* (frequencies, solvers, pattern words, the order of the key; no plaintext and no existing key); *key recovered from adjacent plaintext* (a partial decipherment between the lines or in the margin, clear passages in the same letter, a deciphered copy in the same file); and *key recovered from external plaintext* (a printed edition, a calendar of state papers, a copy in another archive). Two apply a key that already existed: *external key matched* (a key for another correspondent or series, or a key sheet filed elsewhere, found to fit) and *known key applied* (a key already tied to the document or its correspondence). One transcribes: *existing decipherment* (a complete decipherment already existed; the work was transcribing the cipher and checking the decipherment against it). The last two are *not solved* and *not applicable* (not a cipher, or the item could not be obtained). Applying a known key is not mechanical when the key is polyphonic or variable-length, and the profile records such cases in a separate contribution field.

Extent. How much of the text was obtained: complete or partial. The field has no agreed threshold. DECODE marks each record decrypted, partly decrypted or non-decrypted, with the status "freely decided by the owner of the record" (Megyesi et al., 2019), and we met records marked partly decrypted that had been printed in full a century earlier. Computational work reports token accuracy and calls about 95–98 % solved without fixing a cut-off (Ravi and Knight, 2011). Historians call a letter deciphered when the running text reads and a few names or code groups stay open. We state our own rule. A target is *complete* when (1) at least 95 % of its cipher tokens give sense, counted from the reading file; (2) no remaining gap was left untried; (3) every document in it is covered, so that a group with one unread letter is partial; and (4) what stays open is scattered code groups or names, or pieces blocked from outside (no key material, too little text, illegible, or held only in the archive). Conditions (2) and (4) are ours. Open nomenclator codes are counted separately, since a letter can read at 99 % of tokens with many code entries unknown. Where the share of tokens with a value and the share that reads as sense differ, both are given.

The profiles were written by model agents instructed to source every value from the target's folder and to measure lengths with the common script; Section 9 reports how accurate they are.

4 Working method

The record. The agents kept a git repository with one folder per target. Of the 483 targets, 482 have a notes file that the agent maintained while working: what was fetched and from where, the transcription conventions, each hypothesis and solver run with its result, the controls, the literature searched, the reading, and what remains open with the reason. Solver code is in the folder for 305 targets; transcriptions and reading files sit beside it, with crops of the manuscript images where their terms allow it. A finished target was also written up for a public web page and entered in shared ledgers of results. The agents committed their work as they went, and 3,504 of the 4,053 commits name the Claude model that made them in a co-author line (the rest are merges, a few housekeeping commits by hand, and commits from the Codex harness, whose model is recorded in the profile instead), so the history fixes the order and date of the steps. The profiles of Section 3 were filled from these folders and this history.

Models and harness. The work was done by LLMs running as coding agents in terminal harnesses with file, shell and web access. Claude models (mainly Opus 5, Opus 5.5 and Fable 5.1) worked on 449 targets in Claude Code; OpenAI GPT-5 and GPT-6 models worked on 24 targets in the Codex harness, 9 of them shared with a Claude model; the model is not recorded for 19. Long tasks were split among parallel sub-agents, for example one per manuscript page for transcription, and merged by the main agent. The corpus is too unbalanced between model families to compare them, and we do not.

What the humans did. The first author's part was small and mostly at the project level. The first author obtained a DECODE login with the privileges needed to see images and keys and

handed the agent the session cookie, so that it could fetch records itself; set up the harness; and posed each problem as a standing goal, which keeps the agent working until the stated condition holds. During the work the first author occasionally chose which of several leads the agent had identified to pursue. The first author did not transcribe, propose keys or readings, or write code. The project site also takes solutions from outside contributors; those results are not analysed here, and the figures below describe the project's own work. On 28 targets the work was shared between the project and an outside solver (mixed provenance); they stay in the figures and are marked † in the tables and the appendix: astrea1799 (not applicable); biermann1675 (external key matched, partial); comenius1655 (existing decipherment, complete); dedem1788 (existing decipherment, partial); duben1764 (not solved); esp318no95 (key from ciphertext only, partial); foix1563 (external key matched); gilino1527 (external key matched, partial); huangxing (known key applied, complete); ibarra1593 (known key applied); jantini1517 (not solved); lauriere1593 (external key matched, partial); lorraine1592 (key from ciphertext only); malsburg1637 (key from ciphertext only, partial); matignon1586 (known key applied, partial); mercy1648 (key from ciphertext only, partial); michell1751 (existing decipherment, complete); mornay1593 (known key applied, partial); ormanetto1573 (external key matched, partial); ormanetto1576 (external key matched, complete); poyntz1527 (existing decipherment, complete); raince (known key applied, partial); ranzo1528 (key from ciphertext only, partial); revol1593 (known key applied); sessa1593 (known key applied, partial); siena18 (key from ciphertext only, complete); sormano (known key applied, partial); vanreede1793 (not solved). For these the profile records the agent's steps, and the outside solver's part is credited to them in the ledger. In 4 targets the first author went further and supplied a specific source (a printed book the agent could not download) or suggested a route (a candidate key, archive addresses); these are marked in the data. The agents did the rest: locating images through archive interfaces, transcription, statistics, hypotheses, writing and running solvers, searching printed editions, and writing up.

Tools. Solvers were written per target, mostly in Python with C# for speed: hill climbing and simulated annealing over homophonic and syllabic keys, annealers constrained to an alphabetical key design, monotone alignment of ordered key slots, hard EM aligners from ciphertext to a contemporary decipherment, and SAT encodings. The scoring models were character n-gram models built from period corpora, for example nineteenth-century editions of sixteenth-century French correspondence, shared across targets through one registry.

outcome method	n	complete	partial
key from ciphertext only	43	16	21
key from external plaintext	12	6	6
key from adjacent plaintext	47	18	27
key recovered, all routes	*102*	*40*	*62*
external key matched	26	8	13
known key applied	88	42	36
existing decipherment	104	89	14
not solved	114	–	–
not applicable	49	–	–
all targets	483	179	117

Table 1: Outcome method and extent for the 483 targets. The first three methods recover the key here; the next two apply a key that already existed; the sixth transcribes and checks a decipherment that already existed. Complete means the read bar of Section 3 is met.

Conventions. Three project rules bear on the numbers. A negative result is reported only alongside a matched control: a synthetic ciphertext of the same length, alphabet and design that the same solver recovers, or fails to recover in the same way. Every reading is graded by its source: a primary key, a known-plaintext letter, uncertain, or inferred from context or from the alphabetical position of a sign. And before a target was treated as unsolved, the nineteenth-century editions of the correspondence and the DECODE records of the same volume were searched.

5 The corpus

The corpus is restricted to historical (classical) ciphers dated between 1350 and 1950, from any region; the project's work on later or modern material is not reported here. Table 2 summarises the 483 project targets. They hold 1117 ciphertext documents, from 1398 to 1945. The collection is not limited to one continent, but it is weighted toward one. Most targets are diplomatic letters of the sixteenth (183) and seventeenth (93) centuries

century	*n*	plaintext language	*n*
14th c.	3	French	120
15th c.	34	English	88
16th c.	183	Italian	80
17th c.	93	German	48
18th c.	94	Spanish	43
19th c.	28	Latin	34
20th c.	28	Dutch	14
undated	20	Polish	9

Table 2: The corpus by century of the earliest document and by plaintext language (a target with letters in two languages counts under both; 17 languages in all).

region (most documents)	*n*	recovered	applied	existed	not solved
Europe	390	90	106	96	98
North America	16	6	1	5	4
East Asia	2	1	1	0	0
Middle East and South Asia	3	1	0	2	0
not recorded	23	4	6	1	12
all	434	102	114	104	114

Table 3: Outcome by region of the correspondence: the region named by most documents of the target (Europe where no other is named). Columns as in Table 4.

from European chanceries: French, Spanish, Italian, English, Imperial and papal embassy traffic. Outside Europe the project targets are 24 from North America (American Revolutionary, early-republic, Civil War and First World War correspondence), 3 from Latin America, 3 from East Asia and 3 from the Middle East and South Asia (Table 3). The region is recorded for the country field of each document and defaults to Europe where no other country is named; it is unrecorded for 25 project targets. The dating ranges over 34 targets from the fifteenth century and 56 from the nineteenth and twentieth, including telegrams and military radiograms. Another 20 targets have no established date but come from early-modern archival contexts. The main plaintext languages are French 120, English 88, Italian 80, German 48, Spanish 43, Latin 34; 17 languages occur in all. The counts outside Europe are too small to compare regions, and we do not.

A length is known for 354 targets (measured by script on every document for 264); together they hold 1,127,229 enciphered tokens. The agent transcribed 773 of the documents (69 %) itself from archive images; 154 came from DECODE transcriptions and 79 from published ones.

In all, 49 targets are classed not applicable: 16 turned out not to be ciphers (clear text with a cipher instruction, shorthand, key sheets, dummy numbers) and 27 could not be obtained because the manuscript was not digitised or not reachable. The analysis below uses the 434 targets that were worked.

starting point	*n*	recovered	applied	existed	not solved
ciphertext only	110	38	4	4	64
crib from clear text	30	22	1	3	4
decipherment or known plaintext	83	26	6	43	8
a key (archive, print, sibling)	139	14	102	14	9
none recorded	72	2	1	40	29
all	434	102	114	104	114

Table 4: Outcome by what the attack started from (profile field conditions.attack). Columns: key recovered (from ciphertext only, adjacent or external plaintext); key applied (a known key, or an external key matched to the document); a decipherment already existed; not solved. Targets classed not applicable are left out.

cipher family	*n*	recovered	applied	existed	not solved
homophonic or simple + nomenclator	179	45	66	47	21
code or codebook	72	15	9	16	32
undetermined	66	2	3	16	45
syllabic nomenclator	49	10	20	12	7
substitution, no nomenclator	48	24	11	9	4
transposition, polygraphic, polyalphabetic	20	6	5	4	5
all	434	102	114	104	114

Table 5: Outcome by cipher family, from the system types in the profile. Columns as in Table 4.

6 Results

Overall. Table 1 gives the outcome for every target. The text was obtained for 320 of the 434 worked targets: 179 complete and 117 partial; the remaining 114 were not solved. The methods behind the 320 differ widely. The key was recovered here for 102 targets (40 complete): 43 from the ciphertext alone, 47 from adjacent plaintext and 12 from external plaintext. For 114 targets an existing key was applied: 88 with a key already tied to the correspondence and 26 with a key found elsewhere and matched to the document. For 104 targets a complete decipherment already existed, and the contribution was a transcription, a check of the decipherment, or a correction of the catalogue. Counting only the first three methods as cryptanalysis, the model recovered 102 keys in 434 attempts.

Starting point dominates. Table 4 sorts the worked targets by what the attack started from. With a key in hand, from an archive, a printed edition or a sibling letter of the same office, the text was obtained in 130 of 139 cases (94 %). With a

prior reading	n	recovered	applied	existed	not solved
none known	129	41	23	0	65
existed, known before the attempt	151	24	51	62	14
existed, found during the attempt	100	21	30	39	10
existed, found after or not at all	20	11	6	1	2
existed, when found unknown	3	0	1	2	0
unknown	31	5	3	0	23
all	434	102	114	104	114

Table 6: Outcome by whether a reading of the text existed anywhere before this project, and when it was found (the contamination question). Columns as in Table 4.

ciphertext tokens	n	recovered	applied	existed	not solved
< 250	113	23	22	25	43
250–999	119	27	34	19	39
1,000–4,999	89	33	29	16	11
$\geq 5{,}000$	31	12	14	4	1
unknown	82	7	15	40	20
all	434	102	114	104	114

Table 7: Outcome by ciphertext length (enciphered tokens summed over the documents of a target). Columns as in Table 4.

contemporary decipherment of some of the letters, it was obtained in 75 of 83; most of these rebuilt the key from the decipherment and applied it to the undeciphered letters. With a crib from the clear text around the cipher, it was obtained in 26 of 30. From the ciphertext alone, the model recovered the key in 37 of 110 attempts (34 %); in 4 more, a decipherment turned up during the attempt, and 64 were not solved. The other 6 ciphertext-only recoveries began from a crib guessed from the subject of the letter or from the plaintext vocabulary of a sibling key, which uses no plaintext of the document itself.

The ciphertext-only recoveries. Table 8 lists the 43 targets whose key was recovered from the ciphertext alone. Most are homophonic or simple substitutions, some with a small nomenclator, on a few hundred to a few thousand tokens. The key order is known for 10 of them to be alphabetical or partly alphabetical and for 10 to be random; an alphabetical key let the model predict unseen values and confirm them in the text, and in one case (Warsaw 1627) the key's alphabetical order, which the solver did not reward, was the independent check. All but 1 of the 43 carry a verification beyond the reading itself: a matched control, key consistency, or historical consistency with the known events. Of the 43, 18 had a reading somewhere that was found after the key was rebuilt or that covered only part of the target (3 known before the attempt, covering other letters or passages than those whose key was rebuilt); they are independent re-solutions and are marked as such in the data.

Prior readings. Table 6 splits the worked targets by whether a reading existed anywhere before the project, and when it was found. Where none is known (129 targets), the text was obtained for 64, 28 of them complete: 41 by recovering the key (25 from the ciphertext alone) and 23 by applying a key that had never been used on these letters. This is the subset free of contamination, since the model cannot have memorised a decipherment that does not exist. Where a reading existed, it was most often found by the agent during the attempt (100 targets) rather than known beforehand (151); 67 targets had a reading in print, usually in a nineteenth-century edition that the modern lists of unsolved ciphers had missed.

Cipher family matters less. By family (Table 5), homophonic ciphers with nomenclators are the largest group and were obtained most often, but most of those attacks started from a key. Simple and homophonic substitutions without a nomenclator have the most key recoveries relative to their number. The *undetermined* row, targets whose system was never established, is mostly not solved by construction. The comparison across families is confounded by starting point, and the cells are small.

Length. Short ciphertexts (under 250 tokens) were most often not solved (Table 7); above that, the share not solved falls with length, and almost no text of 5,000 tokens or more went unsolved. Long texts are mostly nomenclator letters for which a key was found or rebuilt from sibling decipherments.

Effort. The profiles hold 5,874 solution steps, of which 1313 (22 %) failed or were ruled out. There are 380 solver runs, of which 249 failed or were ruled out, 644 literature searches and 622 transcription steps. Matched controls appear in 169 targets. The balance of steps reflects the work: searching the literature and the archives was as frequent as running solvers, and more often decisive.

What else the work added. Beyond the text, the profiles record corrections to the catalogues (203 targets: a wrong sender, date, or decryption status in DECODE or on a list of unsolved ciphers),

target	year	language	system	tokens	key order	extent
siena14	1420	Italian	homophonic or simple + nomenclator	1291	random	none
r1854	1428	Latin	homophonic or simple + nomenclator	1802	not applicable	complete
cattanei1430	1430	Latin	substitution, no nomenclator	1736	not applicable	partial
siena25	1450	Italian	substitution, no nomenclator	496	random	complete
siena18†	1496	Latin	homophonic or simple + nomenclator	1469	random	complete
esp318no95†	1497	Spanish	homophonic or simple + nomenclator	1364	unknown	partial
vich1515	1515	Spanish	homophonic or simple + nomenclator	3177	unknown	none
soria1523	1523	Spanish	homophonic or simple + nomenclator	–	unknown	partial
ranzo1528†	1528	Italian	code or codebook	8224	random	partial
rangone1530	1529	Italian	substitution, no nomenclator	264	unknown	partial
joachim1530	1530	Italian	homophonic or simple + nomenclator	402	unknown	partial
guiche1551	1551	French	substitution, no nomenclator	431	unknown	partial
memoire1560	1560	French	code or codebook	1715	unknown	partial
r9424	1583	German	homophonic or simple + nomenclator	913	random	partial
harley1582r8505	1585	French	substitution, no nomenclator	3564	random	none
segur	1585	French	syllabic nomenclator	466	alphabetical	complete
lorraine1592†	1592	French	homophonic or simple + nomenclator	1109	unknown	none
stowe166	1593	English	homophonic or simple + nomenclator	126	unknown	complete
segura1596	1596	Spanish	substitution, no nomenclator	125	unknown	complete
senisteros1605	1605	Spanish	substitution, no nomenclator	1447	unknown	complete
anhalt1610	1610	German	undetermined	242	not applicable	none
ferdinand1619	1619	German	homophonic or simple + nomenclator	4181	partly alphabetical	complete
warsaw	1627	Italian	syllabic nomenclator	465	alphabetical	none
harley286	1635	Latin	substitution, no nomenclator	281	random	partial
marburg1635	1635	German	substitution, no nomenclator	590	unknown	partial
malsburg1637†	1637	French, German	transposition, polygraphic, polyalphabetic	13487	not applicable	partial
transylvania1645	1645	Hungarian	undetermined	4785	unknown	partial
mercy1648†	1648	Spanish	homophonic or simple + nomenclator	529	random	partial
wesselenyi1664	1664	Latin	homophonic or simple + nomenclator	244	alphabetical	complete
schenck1706	1706	German	homophonic or simple + nomenclator	1181	alphabetical	partial
potocka1714	1715	French, Polish	homophonic or simple + nomenclator	886	partly alphabetical	partial
kauderbach1754	1754	French	homophonic or simple + nomenclator	8512	random	partial
swieten1759	1759	French	homophonic or simple + nomenclator	553	alphabetical	complete
r1892	1795	Dutch, French	substitution, no nomenclator	2067	random	partial
mountflorence1801	1801	English	code or codebook	39	alphabetical	complete
soglia1848	1848	Italian	code or codebook	955	alphabetical	complete
augur1874	1874	English	transposition, polygraphic, polyalphabetic	246	not applicable	partial
sunyatsen	1916	Chinese	transposition, polygraphic, polyalphabetic	177	partly alphabetical	complete
lejava1921	1921	Ru	transposition, polygraphic, polyalphabetic	167	unknown	complete
voikov1921	1921	Ru	transposition, polygraphic, polyalphabetic	615	unknown	complete
egmond	–	French	substitution, no nomenclator	774	unknown	complete
r9416	–	German	substitution, no nomenclator	3313	unknown	partial
sp106r927	–	French	substitution, no nomenclator	1348	unknown	partial

Table 8: The 43 targets whose key was recovered from the ciphertext alone. † solved in part by an outside contributor.

identifications of an unknown sender or date (159), keys extended or corrected (92), and 25 decipherments with a polyphonic or variable-length key where applying a known key required judgement.

7 Case studies

Toledo to Philip II, 1565 (key from adjacent plaintext). A letter of the viceroy of Sicily during the siege of Malta, with 1,937 two-digit figures inline in clear Spanish (Archivo General de Simancas, Estado leg. 1394). The agent worked out the image route of the Spanish archives portal, transcribed the figures, and took two runs next to clear words as cribs (*seiscientos*, *en tiera*). They fixed eleven letters, and the values came out in alphabetical order, so the remaining letters were predicted from their neighbours and confirmed in words. Nonsense in the decode located misread digits, which were corrected on the images. A companion letter of the same day, found by searching the portal, decodes with the same key. No earlier decipherment is known.

Balbases to Fuenmayor, 1677–78 (existing decipherment; key rebuilt from adjacent plaintext). Fourteen Spanish despatches from the peace congress at Nijmegen, ten with a contemporary decipherment in the margin. Eight keys of the same series were tried and none fitted. Two sub-agents transcribed every cipher and margin line; anchors from *n*-grams repeated across letters seeded a hard-EM aligner that maps each token to a substring of the margin text, and the table con-

verged in two rounds. It reproduces the ten margins. Applied to the four letters that carry no margin, the table gives a syllable stream with a few groups still without a value. A reading audit of 22 October counted those four as decoded, not read, so only the ten margined letters, 71 % of the tokens of the group, count as read, and the profile classes the target by the margins and as partial.

Feuquières to Catinat, 1691 (key from external plaintext). A two-part code of 418 groups (160 distinct) in the "petit chiffre" of the governors of Pignerol, deciphered by Étienne Bazeries in 1893 without publishing the table. A matched control of the inferred design showed that annealing could not recover it: wrong keys outscored the true one. The agent then searched the digitised text of the 1819 *Mémoires* of Catinat for runs of numbers and found a letter of Louvois of 6 September 1690 printed in the same code with its translation. Anchoring by hand and propagating between the two letters gave a value to 586 of 601 tokens, and the Feuquières text agrees with Catinat's own memoir of the same days. The reading audit counts only the Louvois letter (183 tokens) as read, since the Feuquières letter, with twelve groups still open, is decoded but its sense is only half established; the target is partial (30 % of tokens read).

Bordeaux to Brienne, 1653 (external key matched). A French despatch intercepted in London: 813 tokens of graphic signs and numbers, three number series marked by a prime, two dots or an overbar. From sibling keys of the same office the agent inferred the design (alphabetical syllabaries in the marked series) and built an annealer with that constraint. It recovered 99 % of matched controls and failed on the real text; controls with 10 % of the diacritics misread failed in the same way (52 % and 9 %). The agent then found the English Deciphering Branch key sheet for this correspondent in a separate volume of the British Library (DECODE R7537). It confirmed the design exactly, and a check against the images made 21 corrections to the transcription, splitting two merged glyphs and restoring lost marks. The key applied to the corrected transcription read the letter except name codes and rare tokens.

Henry of Navarre to Ségur, 1585–86 (key from ciphertext only). Figures in three letters, 466 tokens. Before any value was known, the frequencies of the upper figures taken modulo five fell into blocks of five, the signature of an alphabetical syllabary of consonants by vowels. An annealer constrained to that design, with the clear French around the cipher as fixed context, converged on a key and recovered 96.6 % of matched controls under random keys of the same shape. A sibling key published by Tomokiyo, with different offsets, read a fourth letter; the constrained annealer then re-derived that fourth key blind at 93 % of letter tokens, against 90 % on a matched control.

Ordnungspolizei radiograms, 1942 (not solved). Seven parts of 27 February 1942 in the German police Doppelkasten, a double Playfair in which each pair passes twice through the same two boxes, 970 letters under one key. The agent implemented the system from the German manuals and attacked it with annealing, iterated local search, tempering, relaxation and SAT. All of these failed on synthetic texts of the same length with a known key: the double pass leaves a local search no gradient. A known-plaintext solver it wrote recovers the boxes from 80 to 111 plaintext pairs on synthetic data, but no plaintext of these messages is available.

8 Where the models failed

Transcription. The largest single cause of failure was the transcription, not the cryptanalysis. The agent transcribed every document of 289 worked targets from images and obtained the text for 217 of them, but the Bordeaux case shows how a transcription that looks usable can defeat a correct solver. Controls with injected transcription noise are cheap and diagnostic, and we recommend them as a standard step.

Short texts and large nomenclators. Large nomenclators on a few hundred tokens were beyond reach without a key or a crib, as they are for human analysts. The agent's value here was in finding the key: on the DECODE record of a sibling letter, in the same archive volume, or in a printed edition.

Undetermined systems. Where the system was never identified, the target was almost never solved. These are mostly short texts, from a few lines of figures to a few hundred tokens: an isolated slip, a postscript, a fragment of a series whose other parts are lost.

Double encipherment and hand systems of the twentieth century. The double Playfair, Enigma traffic and most of the other twentieth-century hand systems and codes in the corpus were not solved, while three short transposition and code targets of 1916–1921 had their keys recovered from the ciphertext alone. The agents wrote correct implementations and competent searches for the failures and reported the negative results with controls.

Finding what was already known. Many targets on current lists of unsolved ciphers had been read long ago, most often in a nineteenth-century printed edition, sometimes in a decipherment bound in the next folio. The agent found these by searching the editions and the neighbouring archive records, and we classify them by the method that applied, never as key recoveries.

9 Threats to validity

The profiles are written by the agents. They were compiled by model agents from the project's notes, partly during the work and partly afterwards, and the key order is unknown for 283 targets, usually because the key was never recovered. The human's role is recorded at the project level for most targets, since the notes rarely mention it; lead choices within a session are not itemised. To estimate accuracy, ten profiles were drawn at random (fixed seed) from the first 82 and checked field by field against the notes, the result tables and the ciphertext files by fresh agents that had not written them. Of 249 fields checked, 38 were unknown. Of the 211 with a value, 196 agreed with the sources (93 %), 5 contradicted them and 10 asserted something the sources do not establish. All recorded lengths re-measured exactly. The errors were of three kinds: placeholder values in profiles of targets never obtained; a modern partial reading filed as a plaintext location; and over-confident system fields for ciphers that were never broken. They were corrected, and the schema gained a *margin* value for plaintext location. The audit was done by agents, not by a human, which limits what it shows.

The outcome method. The method was added to every profile after the work, by review agents that read each target's notes. A rule-based proposal computed from the older profile fields (attack class, prior solution, solution steps) agreed with the reviewers' method for 201 of 284 targets (71 %). The proposal flagged 115 targets as ambiguous, typically whether a sibling key had already been tied to the document (key applied) or had to be matched to it (external key), and whether a key came from plaintext beside the cipher or from plaintext elsewhere; 41 of the 83 disagreements fell among those. The boundary between *known key applied* and *external key matched* is the least certain; the boundary between recovering a key and applying one is well supported by the notes.

The extent rule is ours. The 95 % threshold follows computational practice, not a published standard. A numeric share of tokens read is recorded for 279 of the 320 targets with text (202 measured from the reading files, the rest estimated from the notes). Raising the threshold to 98 % would move at most 34 complete targets to partial; lowering it to 90 % would move at most 12 partial targets to complete. Another 51 targets read above 95 % but are partial for another reason, usually a document of the group left unread.

Selection. The targets came from lists of unsolved ciphers, from DECODE records marked non-decrypted, from sweeps of library and archive catalogues and portals, and the agent chose the order of attack. Targets that looked hopeless were attempted briefly or set aside, so the not-solved count mixes serious failures and brief closures.

No fixed budget. Sessions varied in length, model and operator, and the first author supplied logins and in a few cases a source. The record is of what happened, not of a controlled experiment.

Contamination. A model may have seen a printed decipherment in training. The prior-reading field addresses this directly, and the 129 targets with no known prior reading are the clean subset. For the others, the profile records whether the work used the prior reading, and the method records whether the key was rebuilt or applied.

10 Conclusion

On historical ciphers, an agentic LLM behaves less like a solver than like a research assistant that can also write solvers. It reads the manuscript, finds the key or the decipherment in the archive or in print, recognises a crib in the clear text, and exploits structure such as an alphabetical key. Of

the 320 targets whose text it obtained, the key was recovered here for 102, and from the ciphertext alone for 43. Its failures follow the classical limits: short texts, large nomenclators, double encipherment and noisy transcriptions. Classifying outcomes by method keeps these two contributions apart, and we recommend it to anyone reporting decipherments made with such tools. The profile schema, the 483 profiles and the working record behind them (notes, transcriptions, code and commit history) are released with the paper so that later work can check any profile against its sources and add targets and models on the same terms.

References

Nada Aldarrab and Jonathan May. 2021. Can sequence-to-sequence models crack substitution ciphers? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7226–7235.

Nishant Kambhatla, Logan Born, and Anoop Sarkar. 2023. Decipherment as regression: Solving historical substitution ciphers by learning symbol recurrence relations. In *Findings of the Association for Computational Linguistics: EACL 2023*, pages 2136–2152.

George Lasry, Norbert Biermann, and Satoshi Tomokiyo. 2023. Deciphering Mary Stuart's lost letters from 1578–1584. *Cryptologia*, 47(2):101–202.

George Lasry. 2018. *A Methodology for the Cryptanalysis of Classical Ciphers with Search Metaheuristics*. Kassel University Press, Kassel.

George Lasry. 2023. Armand de Bourbon's polyhomophonic cipher – 1649. In *Proceedings of the 6th International Conference on Historical Cryptology (HistoCrypt 2023)*, pages 105–112. Linköping University Electronic Press.

Yu Li, Qizhi Pei, Mengyuan Sun, Honglin Lin, Chenlin Ming, Xin Gao, Jiang Wu, Conghui He, and Lijun Wu. 2025. CipherBank: Exploring the boundary of LLM reasoning capabilities through cryptography challenge. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5929–5965, Vienna, Austria.

Utsav Maskey, Chencheng Zhu, and Usman Naseem. 2025. Benchmarking large language models for cryptanalysis and side-channel vulnerabilities. arXiv:2505.24621.

Beáta Megyesi, Nils Blomqvist, and Eva Pettersson. 2019. The DECODE database: Collection of historical ciphers and keys. In *Proceedings of the 2nd International Conference on Historical Cryptology (HistoCrypt 2019)*, pages 69–78.

Beáta Megyesi, Bernhard Esslinger, Alicia Fornés, Nils Kopal, Benedek Láng, George Lasry, Karl de Leeuw, Eva Pettersson, Arno Wacker, and Michelle Waldispühl. 2020. Decryption of historical manuscripts: the DECRYPT project. *Cryptologia*, 44(6):545–559.

Beáta Megyesi, Justyna Sikora, Filip Fornmark, Michelle Waldispühl, Nils Kopal, and Vasily Mikhalev. 2023. Historical language models in cryptanalysis: Case studies on English and German. In *Proceedings of the 6th International Conference on Historical Cryptology (HistoCrypt 2023)*. Linköping University Electronic Press.

Sujith Ravi and Kevin Knight. 2011. Bayesian inference for Zodiac and other homophonic ciphers. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 239–247.

Klaus Schmeh. 2017. The top 50 unsolved encrypted messages. Klausis Krypto Kolumne, `https://scienceblogs.de/klausis-krypto-kolumne/the-top-50-unsolved-encrypted-messages/`.

Satoshi Tomokiyo. 2026. Unsolved historical ciphers. `https://cryptiana.web.fc2.com/code/unsolved.htm`.

Table 9: Every target in the corpus, by date of the earliest document. Methods: CT key from ciphertext only; EPT key from external plaintext; APT key from adjacent plaintext; EK external key matched; KK known key applied; DEC existing decipherment; NS not solved; NA not applicable. † solved in part by an outside contributor. Full profiles are in the data release.

target	year	language	system	method	extent	tokens
vico1398	1398	–	undetermined	NS	none	–
benedict1399	1399	Catalan, Latin	homophonic or simple + nomenclator	DEC	complete	–
santacruz1399	1399	Latin	undetermined	DEC	complete	–
jafet1400	1400	Latin	undetermined	DEC	complete	–
exea1408	1408	Latin	undetermined	DEC	complete	–
florence1414	1414	–	substitution, no nomenclator	NA	n/a	–
comuel1415	1415	Latin	undetermined	DEC	complete	–
comuel1416	1416	Latin	undetermined	DEC	complete	–
siena14	1420	Italian	homophonic or simple + nomenclator	CT	none	1291
florence1429	1425	Italian	homophonic or simple + nomenclator	KK	none	99
medici1425	1425	Italian	undetermined	NA	n/a	–
heidelberg597	1426	German	substitution, no nomenclator	DEC	complete	–
malatesta1428	1428	Italian, Latin	undetermined	NS	none	262
mutiliana1428	1428	Italian	homophonic or simple + nomenclator	APT	complete	894
r1854	1428	Latin	homophonic or simple + nomenclator	CT	complete	1802
cattanei1430	1430	Latin	substitution, no nomenclator	CT	partial	1736
maria1435	1435	Catalan	undetermined	NS	none	42
siena8	1440	Italian	homophonic or simple + nomenclator	DEC	complete	166
visconti1446	1446	Italian	syllabic nomenclator	EK	partial	16588
barbavara1447	1447	Italian	syllabic nomenclator	DEC	complete	–
visconti1447	1447	Italian	homophonic or simple + nomenclator	EK	complete	722
birago1448	1448	Italian	homophonic or simple + nomenclator	DEC	complete	513
siena25	1450	Italian	substitution, no nomenclator	CT	complete	496
borghesi1456	1456	Italian	homophonic or simple + nomenclator	APT	none	144
knorre1462	1462	German	undetermined	NA	n/a	–
beatrice1482	1482	Italian	undetermined	DEC	complete	–
matthias1482	1482	Latin	homophonic or simple + nomenclator	DEC	complete	330
sadoleto1482	1482	Italian	homophonic or simple + nomenclator	APT	partial	–
beatrice1486	1486	Italian	undetermined	DEC	complete	–
beatrice1486b	1486	Italian	undetermined	DEC	complete	–
buda1489	1489	Italian	syllabic nomenclator	KK	partial	469
hm64	1490	English, Latin	undetermined	NS	none	–
costabili1493	1493	Italian	not a cipher	NA	n/a	–
siena18†	1496	Latin	homophonic or simple + nomenclator	CT	complete	1469
esp318no95†	1497	Spanish	homophonic or simple + nomenclator	CT	partial	1364
federico1497	1497	Italian	homophonic or simple + nomenclator	DEC	complete	–
garcilaso1498	1498	Spanish	homophonic or simple + nomenclator	NS	none	318
esp318	1500	Spanish	homophonic or simple + nomenclator	KK	none	–
ptr4715	1500	–	undetermined	NA	n/a	–
estrada1503	1503	Spanish	homophonic or simple + nomenclator	KK	partial	–
esp318no93	1504	Spanish	homophonic or simple + nomenclator	NS	none	1322
beatrice1505	1505	Italian	not a cipher	NA	n/a	–
puebla1507	1507	Spanish	undetermined	DEC	complete	–
maximiliana1510	1510	Latin	homophonic or simple + nomenclator	DEC	partial	–
vich1511	1511	Spanish	syllabic nomenclator	APT	partial	22536
bonzagni1512	1512	Italian	substitution, no nomenclator	DEC	partial	–
vich1515	1515	Spanish	homophonic or simple + nomenclator	CT	none	3177
ayala1516	1516	Spanish	homophonic or simple + nomenclator	KK	complete	–
schiner1516	1516	Latin	undetermined	DEC	complete	–
jantini1517†	1517	Italian	undetermined	NS	none	581
r4349	1517	Latin, Swedish	substitution, no nomenclator	KK	partial	131
spinelly1516	1517	English	homophonic or simple + nomenclator	APT	partial	43
caprile1519	1519	Italian	syllabic nomenclator	NS	none	183
caprile1520	1520	Italian	substitution, no nomenclator	EPT	partial	2247
cistarelli1520	1520	Italian	substitution, no nomenclator	NS	none	708
adrian1521	1521	Spanish	syllabic nomenclator	EPT	complete	–
lopehurtado	1522	Spanish	homophonic or simple + nomenclator	APT	partial	2125
poupet1522	1522	French	substitution, no nomenclator	DEC	complete	–
r9658	1522	Spanish	homophonic or simple + nomenclator	DEC	complete	–
sanchez1522	1522	Spanish	homophonic or simple + nomenclator	DEC	partial	2961
frederik1523	1523	German	homophonic or simple + nomenclator	DEC	complete	308
harley6345	1523	English	homophonic or simple + nomenclator	NA	n/a	78
lopehurtado1523	1523	Spanish	homophonic or simple + nomenclator	APT	partial	–
sanchez1523	1523	Spanish	homophonic or simple + nomenclator	KK	partial	–
sessa1523	1523	Spanish	homophonic or simple + nomenclator	EK	partial	–
soria1523	1523	Spanish	homophonic or simple + nomenclator	CT	partial	–
moncada1524	1524	Spanish	homophonic or simple + nomenclator	EPT	complete	442
sessa1524	1524	Spanish	homophonic or simple + nomenclator	APT	partial	6755
ehingen1525	1525	–	undetermined	NA	n/a	–
mellon29	1525	–	undetermined	NS	none	234
bayard1526	1526	French	undetermined	NS	none	56
ghinucci1527	1526	Latin	syllabic nomenclator	EPT	complete	1659
lee1526	1526	English	homophonic or simple + nomenclator	KK	complete	823
raince†	1526	French	homophonic or simple + nomenclator	KK	partial	5629
gilino1527†	1527	Italian	syllabic nomenclator	EK	partial	3111
poyntz1527†	1527	English	homophonic or simple + nomenclator	DEC	complete	44
soria1527	1527	Spanish	homophonic or simple + nomenclator	DEC	complete	–
vasto1527	1527	Spanish	homophonic or simple + nomenclator	KK	complete	1680
dubellay1528	1528	French	homophonic or simple + nomenclator	DEC	complete	–
ranzo1528†	1528	Italian	code or codebook	CT	partial	8224

target	year	language	system	method	extent	tokens
visconti1528	1528	Italian	homophonic or simple + nomenclator	KK	partial	1344
visconti1529	1528	Italian	homophonic or simple + nomenclator	KK	partial	1672
bizozola1520	1529	Italian	substitution, no nomenclator	KK	partial	1914
dubellay	1529	French	homophonic or simple + nomenclator	DEC	complete	194
gramont1529	1529	French	homophonic or simple + nomenclator	KK	complete	4002
kaa4591b	1529	Latin	homophonic or simple + nomenclator	APT	partial	6790
langeac1529	1529	French	substitution, no nomenclator	KK	complete	–
rangone1530	1529	Italian	substitution, no nomenclator	CT	partial	264
sormano†	1529	Italian	homophonic or simple + nomenclator	KK	partial	–
gramont1530	1530	French	homophonic or simple + nomenclator	KK	partial	–
joachim1530	1530	Italian	homophonic or simple + nomenclator	CT	partial	402
isabella1531	1531	Spanish	syllabic nomenclator	DEC	complete	–
mai1531	1531	Spanish	undetermined	NA	n/a	–
muxetula1531	1531	Spanish	homophonic or simple + nomenclator	DEC	complete	–
augurelio1535	1533	German	substitution, no nomenclator	APT	partial	4363
sperantio1534	1533	German, Latin	substitution, no nomenclator	APT	partial	7106
cifuentes1534	1534	Spanish	syllabic nomenclator	DEC	complete	–
kaa4591	1534	German, Latin	homophonic or simple + nomenclator	APT	partial	33462
marechaux1520	1534	French	substitution, no nomenclator	KK	complete	1341
r9367	1535	German	homophonic or simple + nomenclator	KK	partial	534
selve1535	1535	French	homophonic or simple + nomenclator	KK	partial	2524
charles1536	1536	Spanish	syllabic nomenclator	DEC	complete	–
r4248	1536	French	substitution, no nomenclator	DEC	complete	20
rome1536	1536	French	homophonic or simple + nomenclator	KK	partial	675
caracciolo1537	1537	Italian	syllabic nomenclator	KK	partial	–
wardein1537	1537	German	homophonic or simple + nomenclator	APT	partial	4675
devil1539	1539	–	undetermined	NA	n/a	327
vatican5	1542	Italian	syllabic nomenclator	NS	none	6534
sadler1543	1543	English	undetermined	DEC	complete	–
ibrahim1548	1548	Latin	homophonic or simple + nomenclator	NS	none	572
malvezzi1548	1548	Latin	homophonic or simple + nomenclator	KK	complete	–
guiche1551	1551	French	substitution, no nomenclator	CT	partial	431
vargas1552	1552	Spanish	homophonic or simple + nomenclator	DEC	complete	–
grumbach1564	1553	German	homophonic or simple + nomenclator	KK	partial	692
santacroce1552	1553	Italian	homophonic or simple + nomenclator	DEC	complete	3933
harley1582r8499	1554	English	homophonic or simple + nomenclator	EK	none	3152
renard1554	1554	French	undetermined	DEC	complete	–
r1212	1556	Latin	homophonic or simple + nomenclator	KK	complete	288
noailles1558	1558	French	undetermined	NS	none	–
r1873	1558	Italian	homophonic or simple + nomenclator	DEC	complete	176
burghley1559	1559	English	code or codebook	DEC	complete	12
percy1559	1559	–	undetermined	NS	none	63
throck1559	1559	English	homophonic or simple + nomenclator	KK	none	729
throck1559margin	1559	–	code or codebook	NS	none	222
memoire1560	1560	French	code or codebook	CT	partial	1715
throck1560	1560	English	undetermined	DEC	complete	–
throckmorton	1560	English	homophonic or simple + nomenclator	DEC	complete	–
smith1562	1562	English	homophonic or simple + nomenclator	KK	partial	–
coligny1563	1563	French	homophonic or simple + nomenclator	DEC	complete	–
foix1563†	1563	–	homophonic or simple + nomenclator	EK	none	1007
lomer1563	1563	English	code or codebook	DEC	complete	7
middelmore1563	1563	English	code or codebook	DEC	complete	18
rennes1563	1563	French	homophonic or simple + nomenclator	KK	none	2034
charlesixducroc	1565	French	homophonic or simple + nomenclator	DEC	complete	–
toledo1565	1565	Spanish	substitution, no nomenclator	APT	complete	2087
ducroc	1567	French	undetermined	NS	none	166
norreys1567	1567	English	homophonic or simple + nomenclator	KK	partial	7
nuncio1566	1567	Italian	homophonic or simple + nomenclator	KK	complete	7817
alessandrino1568	1568	Italian	homophonic or simple + nomenclator	KK	partial	10611
moray	1568	–	undetermined	NS	none	134
wod1568	1568	–	undetermined	NS	none	39
hamilton1569	1569	Scots	homophonic or simple + nomenclator	DEC	complete	–
throck1569	1569	English	homophonic or simple + nomenclator	KK	complete	–
norfolk1570	1570	Scots	homophonic or simple + nomenclator	NS	none	–
randolph1570	1570	English	homophonic or simple + nomenclator	DEC	complete	–
birago	1571	Italian	undetermined	NS	none	483
burghley1572	1571	English	homophonic or simple + nomenclator	NS	none	4
r8356	1571	English	homophonic or simple + nomenclator	NS	none	11
mondoucet	1572	French	substitution, no nomenclator	APT	partial	7293
r8358	1572	English	homophonic or simple + nomenclator	NS	none	15
r8361	1572	English	homophonic or simple + nomenclator	NS	none	21
walsingham1572nov	1572	English	undetermined	NA	n/a	–
lanssac	1573	French	homophonic or simple + nomenclator	APT	complete	2165
ormanetto1573†	1573	Italian	homophonic or simple + nomenclator	EK	partial	1324
r8362	1573	English	homophonic or simple + nomenclator	NA	n/a	2
r8364	1573	English	homophonic or simple + nomenclator	EPT	partial	26
walsingham1572	1573	English	substitution, no nomenclator	NS	none	12
nevers1574	1574	French, Italian	substitution, no nomenclator	DEC	partial	–
philip1574	1574	Spanish	homophonic or simple + nomenclator	DEC	complete	–
chodkiewicz1575	1575	German	substitution, no nomenclator	DEC	complete	–
fulda1576	1576	Latin	homophonic or simple + nomenclator	KK	partial	270
ormanetto1576†	1576	Italian	homophonic or simple + nomenclator	EK	complete	389
ungnad1576	1576	–	homophonic or simple + nomenclator	NA	n/a	–
sinzendorff1578	1578	–	undetermined	NA	n/a	–
dandini1580	1580	Italian	homophonic or simple + nomenclator	DEC	complete	509

target	year	language	system	method	extent	tokens
dandini1580r73	1580	Italian	homophonic or simple + nomenclator	DEC	partial	1082
sauli1579	1580	Italian	homophonic or simple + nomenclator	DEC	complete	2392
r4157	1581	French	homophonic or simple + nomenclator	KK	complete	289
walsingham1581	1581	English	homophonic or simple + nomenclator	DEC	complete	562
r9319	1583	De	undetermined	DEC	complete	–
r9424	1583	German	homophonic or simple + nomenclator	CT	partial	913
segur1583	1583	French	syllabic nomenclator	NS	none	58
fleury1585	1584	French	substitution, no nomenclator	NS	none	276
lodi1584	1584	Spanish	homophonic or simple + nomenclator	DEC	complete	307
harley1582r8505	1585	French	substitution, no nomenclator	CT	none	3564
matignon1586†	1585	French	homophonic or simple + nomenclator	KK	partial	17913
segur	1585	French	syllabic nomenclator	CT	complete	466
walsingham1585	1585	English	homophonic or simple + nomenclator	EPT	partial	–
wotton1585	1585	English	homophonic or simple + nomenclator	APT	complete	2866
bergamo1585	1586	Italian	homophonic or simple + nomenclator	KK	complete	3216
clair357	1586	French	homophonic or simple + nomenclator	DEC	partial	–
harley1582r8500	1586	English	homophonic or simple + nomenclator	APT	partial	61
harley1582r8504	1586	English	homophonic or simple + nomenclator	APT	complete	7
navarre1586	1586	French	syllabic nomenclator	KK	partial	108
stafford1586	1586	English	homophonic or simple + nomenclator	DEC	complete	28
guise1587	1587	French	homophonic or simple + nomenclator	KK	none	2484
mercoeur1587	1587	French	homophonic or simple + nomenclator	KK	partial	952
morosini1588	1587	Italian	homophonic or simple + nomenclator	KK	partial	238550
needham1587	1587	English	substitution, no nomenclator	APT	complete	227
nevers1587	1587	French	homophonic or simple + nomenclator	EK	complete	27
vieuville1587	1587	French	homophonic or simple + nomenclator	EK	none	–
cobham1588	1588	English	homophonic or simple + nomenclator	APT	none	–
harley287	1588	English	homophonic or simple + nomenclator	APT	partial	–
nevers1588	1588	French	homophonic or simple + nomenclator	APT	complete	476
r3708	1588	French	homophonic or simple + nomenclator	APT	complete	336
mendoza1589	1589	Spanish	syllabic nomenclator	DEC	complete	822
morosini1589	1589	Italian	syllabic nomenclator	DEC	complete	1448
orbais	1589	French	homophonic or simple + nomenclator	KK	partial	91
ottobon	1589	Italian	syllabic nomenclator	EK	none	1528
r2276	1589	French	homophonic or simple + nomenclator	KK	partial	3920
bienvenut1590	1590	French	homophonic or simple + nomenclator	KK	complete	26
champagne1590	1590	French	homophonic or simple + nomenclator	EK	partial	110
gonzaga1590	1590	Italian	homophonic or simple + nomenclator	KK	complete	926
lorraine1592†	1592	French	homophonic or simple + nomenclator	CT	none	1109
pelissier1592	1592	French	homophonic or simple + nomenclator	KK	partial	18451
ibarra1593†	1593	Spanish	syllabic nomenclator	KK	none	598
lauriere1593†	1593	French	syllabic nomenclator	EK	partial	667
lebel1593	1593	French	homophonic or simple + nomenclator	KK	partial	1935
mauclerc1593	1593	French	undetermined	DEC	complete	–
mornay1593†	1593	French	undetermined	KK	partial	278
nevers1593	1593	French	homophonic or simple + nomenclator	KK	none	395
revol1593†	1593	French	syllabic nomenclator	KK	none	86
sega1593	1593	Italian	homophonic or simple + nomenclator	NS	none	–
sessa1593†	1593	Spanish	syllabic nomenclator	KK	partial	1310
stowe166	1593	English	homophonic or simple + nomenclator	CT	complete	126
casimir1618	1594	Dutch	undetermined	NS	none	259
joyeuse	1594	French	syllabic nomenclator	NS	none	358
nevers1595	1595	French	homophonic or simple + nomenclator	NS	none	753
sloane3188	1595	English	not a cipher	NA	n/a	–
segura1596	1596	Spanish	substitution, no nomenclator	CT	complete	125
bethune	1601	French	syllabic nomenclator	APT	partial	2849
breves1603	1603	French	homophonic or simple + nomenclator	DEC	complete	–
sirtori1604	1604	–	undetermined	NS	none	747
senisteros1605	1605	Spanish	substitution, no nomenclator	CT	complete	1447
anhalt1610	1610	German	undetermined	CT	none	242
brulart1615	1615	French	homophonic or simple + nomenclator	DEC	partial	–
andreae1616	1616	German	substitution, no nomenclator	APT	complete	156
cocquet	1616	French	undetermined	NS	none	–
r4817	1618	German	substitution, no nomenclator	EPT	complete	79
ferdinand1619	1619	German	homophonic or simple + nomenclator	CT	complete	4181
haga1620	1620	Dutch	homophonic or simple + nomenclator	KK	complete	–
sp106box10	1623	Italian	code or codebook	KK	none	–
sp106r704	1623	Italian	syllabic nomenclator	NS	none	–
sp106r725	1623	Italian	code or codebook	NS	none	–
damiata1624	1625	Italian	homophonic or simple + nomenclator	DEC	complete	361
sp106r722	1625	English, Italian	code or codebook	NS	none	420
warsaw	1627	Italian	syllabic nomenclator	CT	none	465
boswell1628	1628	English	substitution, no nomenclator	KK	complete	442
kemeny1628	1628	–	undetermined	NS	none	–
r1862	1628	Italian	homophonic or simple + nomenclator	EPT	complete	1280
r1874	1628	Italian	homophonic or simple + nomenclator	DEC	complete	2245
riksarkivet4284	1628	Latin	undetermined	EK	none	822
rusdorff1628	1628	German, Latin	undetermined	NS	none	–
pallotto1629	1629	Italian	homophonic or simple + nomenclator	DEC	complete	118931
richelieu	1629	French	homophonic or simple + nomenclator	APT	complete	421
sg1629	1629	–	undetermined	NA	n/a	–
riksarkivet1628	1631	German	homophonic or simple + nomenclator	APT	partial	1130
ceva1632	1632	Italian	syllabic nomenclator	DEC	complete	2400
oxenstierna1633	1633	Swedish	homophonic or simple + nomenclator	NS	none	–
ferdinand1634	1634	German	homophonic or simple + nomenclator	NS	none	–

target	year	language	system	method	extent	tokens
ferdinand3	1634	Latin	syllabic nomenclator	APT	partial	1719
ormonde	1634	English	homophonic or simple + nomenclator	APT	partial	59
harley286	1635	Latin	substitution, no nomenclator	CT	partial	281
koerler1675	1635	German	undetermined	NA	n/a	–
marburg1635	1635	German	substitution, no nomenclator	CT	partial	590
emery1636	1636	French	undetermined	NS	none	–
rohan1636	1636	French	undetermined	NS	none	–
heusner1637	1637	German	homophonic or simple + nomenclator	KK	complete	259
malsburg1637†	1637	French, German	transposition, polygraphic, polyalphabetic	CT	partial	13487
kurtz1639	1639	German	substitution, no nomenclator	APT	partial	22913
baner1640	1640	German	homophonic or simple + nomenclator	APT	partial	288
harley7001	1641	English	homophonic or simple + nomenclator	EK	partial	254
joao1642	1642	Portuguese	homophonic or simple + nomenclator	APT	partial	1817
boswell	1643	English	homophonic or simple + nomenclator	KK	complete	457
forster	1644	French	substitution, no nomenclator	KK	complete	207
lucca	1644	Italian	substitution, no nomenclator	EPT	partial	231
goring1645	1645	English	substitution, no nomenclator	KK	complete	338
hm1645	1645	English	homophonic or simple + nomenclator	DEC	complete	3373
rupert	1645	English	homophonic or simple + nomenclator	KK	partial	1412
rupert1645	1645	English	homophonic or simple + nomenclator	EK	complete	304
sp106r787	1645	French	homophonic or simple + nomenclator	DEC	complete	1569
sp106r931	1645	English	code or codebook	NS	none	–
transylvania1645	1645	Hungarian	undetermined	CT	partial	4785
sp106r924	1646	English	homophonic or simple + nomenclator	DEC	complete	2012
sp106r929	1646	English	homophonic or simple + nomenclator	KK	complete	348
charlesi	1648	English	undetermined	NS	none	200
mercy1648†	1648	Spanish	homophonic or simple + nomenclator	CT	partial	529
r4941	1648	–	undetermined	NS	none	–
conti1649	1649	French	substitution, no nomenclator	KK	complete	1692
laigue1649	1649	French	syllabic nomenclator	APT	complete	3367
hamilton	1650	English	homophonic or simple + nomenclator	NA	n/a	102
wallis1653	1651	English, French	syllabic nomenclator	DEC	complete	480
conley1652	1652	English	homophonic or simple + nomenclator	NS	none	23
urquhart	1652	English	code or codebook	KK	partial	349
bordeaux	1653	French	syllabic nomenclator	EK	complete	813
thurloe	1653	–	homophonic or simple + nomenclator	NS	none	136
comenius1655†	1655	Latin	substitution, no nomenclator	DEC	complete	88
copinger1656	1656	English	undetermined	NS	none	16
dugard1656	1656	–	undetermined	NS	none	27
waddall1656	1656	English	homophonic or simple + nomenclator	NS	none	50
letellier	1657	French	undetermined	NS	none	202
rome1658	1658	Italian	undetermined	NA	n/a	–
hyde	1659	–	not a cipher	NA	n/a	41
miranda1660	1660	French, Italian, Latin, Portuguese, Spanish	undetermined	NA	n/a	–
r688	1664	Serbian	undetermined	NA	n/a	240
wesselenyi1664	1664	Latin	homophonic or simple + nomenclator	CT	complete	244
erbach1653	1666	German, Latin	substitution, no nomenclator	APT	complete	610
perwich	1670	English	transposition, polygraphic, polyalphabetic	DEC	complete	494
hesse1672	1672	German	syllabic nomenclator	EK	partial	40
hernannunez1674	1674	Spanish	syllabic nomenclator	KK	complete	1716
biermann1675†	1675	French	syllabic nomenclator	EK	partial	263
colbert	1675	French	undetermined	NS	none	106
balbases1677	1677	Spanish	syllabic nomenclator	DEC	partial	–
carpio1677	1677	Spanish	syllabic nomenclator	APT	complete	3504
ronquillo1676	1677	Spanish	syllabic nomenclator	DEC	partial	23463
ormond1678	1678	English	code or codebook	NS	none	20
r1871	1680	Italian	homophonic or simple + nomenclator	DEC	complete	610
szembek1689	1689	Latin	homophonic or simple + nomenclator	DEC	complete	439
chaulnes	1690	French	syllabic nomenclator	NS	none	311
feuquieres	1690	French	code or codebook	EPT	partial	601
catinat1691	1691	French	code or codebook	KK	complete	12439
ilten1697	1697	French	code or codebook	DEC	complete	74
yard1699	1699	English	code or codebook	KK	complete	368
catinat1702	1702	French	code or codebook	NS	none	591
prussia1702	1702	French	undetermined	NA	n/a	56
stepney	1702	English	code or codebook	NS	none	24
ilten1705	1705	French, German	code or codebook	DEC	complete	418
bay1706	1706	Hungarian	syllabic nomenclator	EK	partial	414
papai1706	1706	Hungarian	syllabic nomenclator	KK	partial	16861
rakoczi1704	1706	French	syllabic nomenclator	EK	partial	2875
schenck1706	1706	German	homophonic or simple + nomenclator	CT	partial	1181
vetesi1705	1706	Hungarian	syllabic nomenclator	EK	complete	–
henter1707	1707	Hungarian	homophonic or simple + nomenclator	DEC	complete	–
r7475	1707	Polish	homophonic or simple + nomenclator	DEC	complete	2444
rakoczi1707	1707	French	syllabic nomenclator	EK	complete	1108
sieniawski1708	1709	Polish	homophonic or simple + nomenclator	DEC	complete	–
turculet1709	1709	Polish	homophonic or simple + nomenclator	DEC	complete	877
desmarets1710	1710	French	homophonic or simple + nomenclator	KK	partial	471
geertruidenberg	1710	French	code or codebook	NS	none	690
papai1710	1710	Hungarian	undetermined	NS	none	371
villars1710	1710	French	code or codebook	NS	none	173
rakoczi1711	1711	Latin	syllabic nomenclator	DEC	partial	2531
anne1711	1712	English	code or codebook	DEC	complete	414
lagau1712	1712	French	undetermined	NA	n/a	–
chomentowski1713	1713	Polish	syllabic nomenclator	DEC	partial	2246

target	year	language	system	method	extent	tokens
mniszech1714	1714	Latin, Polish	undetermined	NS	none	232
stenbock1713	1714	–	undetermined	NA	n/a	–
potocka1714	1715	French, Polish	homophonic or simple + nomenclator	CT	partial	886
goertz1717	1717	German	homophonic or simple + nomenclator	NS	none	809
duran1718	1718	–	code or codebook	NS	none	317
nunzio1718	1718	Italian	homophonic or simple + nomenclator	KK	complete	34243
bl32305	1719	French	code or codebook	NS	none	354
windischgraetz1720	1720	German	syllabic nomenclator	KK	complete	223
windischgraetz1720b	1720	German	homophonic or simple + nomenclator	KK	complete	192
colonia1721	1721	Italian	homophonic or simple + nomenclator	KK	complete	189
windischgraetz1721	1721	German	homophonic or simple + nomenclator	KK	partial	181
palm1727	1727	German	homophonic or simple + nomenclator	NS	none	185
visconti1727	1727	Italian	homophonic or simple + nomenclator	NA	n/a	193
r7707	1728	–	undetermined	NS	none	205
portstemarie1729	1729	French	code or codebook	APT	complete	141
sp106r597	1731	French	homophonic or simple + nomenclator	NA	n/a	43
august1733	1733	–	undetermined	NA	n/a	–
mecklenburg1734	1734	German	code or codebook	NS	none	598
nunzio1736	1736	Italian	homophonic or simple + nomenclator	KK	partial	7386
carre1742	1742	–	code or codebook	NS	none	1323
ilten1743	1743	French	code or codebook	EK	complete	223
esterhazy1744	1744	French	code or codebook	NS	none	176
waldeck1744	1744	German	homophonic or simple + nomenclator	NS	none	717
sp95box136	1748	–	undetermined	NA	n/a	–
michell1751†	1751	French	code or codebook	DEC	complete	550
hellen1752	1752	French	code or codebook	NS	none	2651
jones1753	1753	English	not a cipher	NA	n/a	–
kauderbach1754	1754	French	homophonic or simple + nomenclator	CT	partial	8512
affry1757	1757	French	homophonic or simple + nomenclator	APT	partial	8980
affry1757c	1757	French	homophonic or simple + nomenclator	NS	none	761
acciaiuoli1757	1758	Italian	homophonic or simple + nomenclator	KK	complete	8326
eichel1758	1758	–	homophonic or simple + nomenclator	NA	n/a	139
swieten1757	1758	French	homophonic or simple + nomenclator	DEC	complete	89
orvilliers1759	1759	–	undetermined	NA	n/a	–
swieten1759	1759	French	homophonic or simple + nomenclator	CT	complete	553
mercier1761	1761	French	code or codebook	APT	partial	818
r1875	1761	Italian	substitution, no nomenclator	DEC	complete	1565
duben1764†	1764	French	undetermined	NS	none	4340
lucini1767	1767	Italian	homophonic or simple + nomenclator	KK	complete	152
smallwood1777	1777	English	not a cipher	NA	n/a	5
martinique1778	1778	French	code or codebook	APT	partial	285
destaing	1779	French	code or codebook	NS	none	217
pcc1780	1780	English	code or codebook	DEC	complete	–
balfour1781	1781	English	transposition, polygraphic, polyalphabetic	DEC	complete	6337
livingston1781	1781	English	homophonic or simple + nomenclator	DEC	complete	17
deswart1782	1782	Dutch	code or codebook	EK	partial	9603
schoning1783	1783	French	code or codebook	APT	complete	4100
rechteren1785	1785	Dutch	homophonic or simple + nomenclator	DEC	complete	424
jefferson1787	1786	English	code or codebook	APT	partial	218
vanreede1787	1787	French	code or codebook	KK	complete	471
deboli1790	1788	Polish	code or codebook	DEC	complete	211
dedem1788†	1788	Dutch	code or codebook	DEC	partial	941
legation1788	1788	English	code or codebook	DEC	complete	84
deboli1789	1789	Polish	syllabic nomenclator	APT	partial	29465
reede1789	1789	French	code or codebook	NS	none	4296
deboli1791	1791	Polish	code or codebook	DEC	partial	3082
jablonowski1791	1791	French	undetermined	NA	n/a	–
vanreede1792	1792	French	code or codebook	KK	complete	83
vanreede1793†	1793	Dutch	code or codebook	NS	none	696
r1892	1795	Dutch, French	substitution, no nomenclator	CT	partial	2067
r2236	1795	Dutch	undetermined	NS	none	–
r2242	1795	Dutch	substitution, no nomenclator	EK	partial	1261
wilhelmina1795	1795	Dutch, French	code or codebook	NS	none	–
wilkinson1789	1795	English	code or codebook	NS	none	3408
hereditary1796	1796	French	code or codebook	APT	complete	881
r2234	1796	French	transposition, polygraphic, polyalphabetic	KK	complete	1321
reede1796	1796	French	code or codebook	NS	none	1512
yvoy1797	1797	Dutch	code or codebook	DEC	complete	3487
astrea1799†	1798	French	undetermined	NA	n/a	–
jones1802	1798	English	code or codebook	EPT	partial	1049
mellon136	1798	German	substitution, no nomenclator	KK	complete	1670
dedem1799	1799	–	code or codebook	NS	none	–
king1801	1801	English	undetermined	DEC	complete	–
mountflorence1801	1801	English	code or codebook	CT	complete	39
murray1801	1801	English	code or codebook	APT	complete	6
r1944	1801	Dutch	code or codebook	DEC	complete	110
r2232	1801	French	substitution, no nomenclator	KK	complete	1065
r1942	1803	Dutch	code or codebook	NS	none	286
toulon1803	1803	Dutch	code or codebook	NS	none	110
fagel1804	1804	French	homophonic or simple + nomenclator	NS	none	128
graham1806	1806	English	code or codebook	NS	none	–
erving1807	1807	English	code or codebook	DEC	complete	581
armstrong	1808	English	code or codebook	EPT	complete	49
spaen1808	1808	–	code or codebook	NS	none	303
roell1809	1809	–	code or codebook	NS	none	2585

target	year	language	system	method	extent	tokens
jqa	1812	English	code or codebook	APT	partial	390
poinsett1813	1813	English	code or codebook	NS	none	560
castelcicala1816	1816	Italian	code or codebook	DEC	partial	9428
loeschnen1816	1816	German	not a cipher	NA	n/a	–
hesse1824	1824	German	transposition, polygraphic, polyalphabetic	APT	complete	164
zeschau1841	1841	French, German	syllabic nomenclator	NS	none	1896
soglia1848	1848	Italian	code or codebook	CT	complete	955
civilwar1862	1862	English	undetermined	NA	n/a	–
eckert1862	1862	English	transposition, polygraphic, polyalphabetic	KK	none	100484
milroy	1862	English	transposition, polygraphic, polyalphabetic	KK	complete	79
union1864	1864	English	transposition, polygraphic, polyalphabetic	DEC	n/a	59
army1865	1865	English	undetermined	NA	n/a	–
vienna1873	1870	English	code or codebook	NS	none	–
augur1874	1874	English	transposition, polygraphic, polyalphabetic	CT	partial	246
catokwacopa	1875	English	transposition, polygraphic, polyalphabetic	NS	none	441
bell1900	1900	English	code or codebook	KK	complete	7
gupta1917	1916	English	undetermined	NA	n/a	–
huangxing[†]	1916	Chinese	transposition, polygraphic, polyalphabetic	KK	complete	275
sunyatsen	1916	Chinese	transposition, polygraphic, polyalphabetic	CT	complete	177
censorship1917	1917	–	undetermined	NA	n/a	–
lima1917	1917	German	undetermined	NA	n/a	–
tampico1917	1917	–	undetermined	NA	n/a	–
adfgvx	1918	German	transposition, polygraphic, polyalphabetic	KK	partial	3966
ludendorff1918	1918	–	undetermined	NS	none	328
stockholm1918	1918	–	code or codebook	NS	none	938
karakhan1921	1921	–	undetermined	NS	none	181
kopp1921	1921	–	undetermined	NS	none	717
lejava1921	1921	Ru	transposition, polygraphic, polyalphabetic	CT	complete	167
stomoniakov1921	1921	–	undetermined	NS	none	1042
voikov1921	1921	Ru	transposition, polygraphic, polyalphabetic	CT	complete	615
voikov1921b	1921	Ru	transposition, polygraphic, polyalphabetic	NS	none	398
roosevelt	1935	–	not a cipher	NA	n/a	52
blume	1937	Spanish	transposition, polygraphic, polyalphabetic	NS	none	775
zhongshan	1938	Chinese	code or codebook	NA	n/a	7
dagapeyeff	1939	–	undetermined	NS	none	196
orpo1942	1942	German	transposition, polygraphic, polyalphabetic	NS	none	968
orpo1942jun	1942	German	transposition, polygraphic, polyalphabetic	DEC	complete	223
powgerman1942	1942	German	undetermined	NS	none	–
censorship	1943	German	substitution, no nomenclator	NS	none	–
abwehr	1944	–	undetermined	NS	none	924
ccr1944	1944	–	undetermined	NA	n/a	–
buenosaires1945	1945	German	undetermined	NA	n/a	–
enigma	1945	German	transposition, polygraphic, polyalphabetic	NS	none	97
blancmesnil	–	French	undetermined	NA	n/a	–
burgo1527	–	Spanish	not a cipher	NA	n/a	–
debosnys	–	–	undetermined	NS	none	975
egmond	–	French	substitution, no nomenclator	CT	complete	774
fr3029	–	French	undetermined	KK	complete	–
fr3029f134	–	French	homophonic or simple + nomenclator	KK	complete	926
fr3092	–	French	homophonic or simple + nomenclator	KK	complete	4521
mercoeur1586	–	French	homophonic or simple + nomenclator	NS	none	217
mid4131829	–	English	undetermined	NA	n/a	–
r1818	–	–	undetermined	NS	none	355
r8870	–	–	code or codebook	NS	none	901
r8871	–	–	undetermined	NS	none	77
r9416	–	German	substitution, no nomenclator	CT	partial	3313
r9586	–	Italian	code or codebook	NS	none	969
siena10	–	Italian	substitution, no nomenclator	DEC	complete	25
siena19	–	Italian	undetermined	NS	none	118
siena21	–	Italian	undetermined	NS	none	107
siena7	–	Italian	undetermined	NS	none	363
sp106r927	–	French	substitution, no nomenclator	CT	partial	1348
sp53	–	–	undetermined	NS	none	1235